UNIT III – MAGNETIC PROPERTIES

Part- A Question and Answers – 2 marks

1.Define the terms Susceptibility and Permeability.

Magnetic Susceptibility (χ_m):

It is defined as the ratio between the magnetic flux density (B) and the magnetic field intensity(H).

$$\chi_m = \frac{I}{H}$$

Magnetic Permeability (μ):

It is defined as the ratio between the magnetic flux density (B) and the magnetic field intensity (H).

$$\mu = \mu_o \mu_r = \frac{B}{H}$$

2. Define Diamagnetic Material?

It is a material in which there is no permanent dipole moment (or) magnetic moment in each atom. The induced magnetic moment produced in these materials during the application of the external magnetic field decreases the magnetic induction present in the specimen.

3.Define Paramagnetic Material?

It is a material in which there is permanent dipole moment (or) magnetic moment in each atom. The induced magnetic moment produced in these materials during the application of the external magnetic field increases the magnetic induction present in the specimen.

4.Define Ferromagnetic Material?

It is a material in which there is enormous permanent dipole moment (or) magnetic moment in each moment in each atom. The induced magnetic moment produced during the application of the external magnetic field is very large and it increases the magnetic induction present in the specimen.

5.What are Diamagnetic materials and mention their properties

The diamagnetism is the phenomenon by which the induced magnetic moment is always in the opposite direction of the applied magnetic field.

The magnetic material having negative susceptibility is called a diamagnetic material.

Further for the diamagnetic materials, each atom has no permanent magnetic moment.

6.What are called Paramagnetic materials and give their Properties?

-If we apply the external magnetic field, there is an enormous magnetic moment along the field direction and the magnetic induction will increase.

-Paramagnetic materials attract magnetic lines of force.

-They possess permanent magnetic dipoles.

7.State Ferromagnetic material and give their properties.

Ferromagnetism is a phenomenon by which spontaneous magnetization occurs when $T \leq \theta$ and so even in the absence of applied field, the magnetic moments are enormous. Here θ is the curie temperature of the material.

Ferromagnetism arises when the exchange energy is favorable for spin alignment.

If a material acquires a relatively high magnetization in a weak field, then it is ferromagnetic.

8.Define Ferrimagnetic materials and mention their properties?

It is a special case of antiferromagnetic in which antiparalled moments are of different magnitudes and a large magnetization arises.

Magnitude of susceptibility is very large and positive.

9.What are called Ferrites?

1. Ferrites are the modified structures of Iron with no carbon and are good examples of Ferrimagnetism in which the spins of adjacent ions in the presence of a magnetic field are in opposite senses and with different magnitudes.

2. They are made from ceramic ferro magnetic compounds.

10.Give the types of Ferrites.

Regular spinel:

In the regular spinel, each trivalent metal ion occupies an octahedral site(B) amd each divalent ion occupies a tetrahedral site(A) of FCC oxygen lattice.

Example: $X^{2+}Fe_2^{3+}O_4$ where X is divalent metal ion such as Mn, Cu, Ni, Mg, etc.

Inverse spinel:

In the inverse spinel, trivalent metal ions occupy all tetrahedral sites (A) and half of octahedral sites(B) and the rest of octahedral sites (B) are occupied by divalent metal ions. Example: $Fe^{3+}(Fe^{2+}Fe+)O_4$=Ferrous ferrite.

11.State some application of Ferrites.

- Ferrites are used to produce low frequency ultrasonic waves by magnetostriction principle. Further these are used in the electromechanical transducers.
- Ferrite rods are used in radio receivers(particularly In medium wave coil) to increase the sensitivity and selectivity of receiver.
- Ferrites like Nickel Zinc Ferrites are used as cores in audio and T.V. transformers.

12.What is called spontaneous magnetisation?

In ferromagnetic materials the molecular magnets in the ferromagnetic material is aligned in such a way that they exhibit a magnetization even in the absence of an external magnetic field. This is called spontaneous magnetization.

13.What do you meant by Domain?

A region in a ferromagnetic material where all the magnetic moments are aligned in the same direction is called a domain. A magnetic domain is completely magnetized and has definite boundaries. So a ferromagnetic material is divided up into these small regions, called domains, each of which is at all times completely magnetized.

14.Define Hysteresis.

The hysteresis of ferromagnetic material refers to the lag of magnetization or magnetic induction behind the magnetizing field.

15.Define Magnetostricition effect.

A specimen when magnetized suddenly experiences a slight change in its length which is due to rearrangement of domains inside. This is called magnetostriction.

16.What is meant by Energy Product?

Energy product of a magnetic material Is the product of residual magnetic induction(Br) and demagnetizing field strength(Hc).

17.What do you meant by Exchange Energy.

It is the energy associated with the quantum mechanical coupling that aligns the individual atomic dipoles with in a single domain. It arises from interaction of electron spins. It depends upon the interatomic distance.

18.Define the terms coercivity and retentivity.

When the external magnetic field is applied to a magnetic material is removed, the magnetic material will not loss its magnetic property immediately there exists some residual intensity of magnetization in the specimen even when the magnetic field is cut off. This is called residual magnetism or retentivity.

The residual magnetism can be completely removed from the material by applying a reverse magnetic field. Hence coercivity of the magnetic material is the strength of reverse magnetic field (-Hc) which is used to completely demagnetize the material.

19.Distinguish between Hard and soft magnetic material.

- Hard magnetic materials are magnetic materials which cannot be easily magnetized and demagnetized.
- They have large hysteresis loss due to large hysteresis loop area.
- These materials have small values for permeability and susceptibility.

Soft magnetic material:

- Soft magnetic materials are magnetic materials which can be easily magnetized and demagnetized.
- They have small hysteresis loss due to small hysteresis loop area.
- These materials have large values for permeability and susceptibility.

UNIT IV – SEMICONDUCTORS AND SUPERCONDUCTORS

Part- A Question and Answers – 2 marks

1.Define Intrinsic Semiconductors.

A semiconductor in which holes and electrons are created exclusively by thermal excitation across the energy gap is called an intrinsic semiconductor. A pure crystal of silicon or germanium is an intrinsic semiconductor.

2.What are p-type and n-type semiconductors?

P-type semiconductor is the one having holes as the majority charge carriers and electrons as the minority charge carriers.

Example: Silicon or Germanium doped with trivalent impurities like Al, Ga and In. N-type semiconductor is the one having electrons as the majority charge carriers and holes as the minority charge carriers.

Example: Silicon or Germanium doped with pentavalent impurities like P, As and Sb.

3.What is the meaning of bandgap of a semiconductor?

Bandgap (or) energy gap of a semiconductor is the region of energies which are not allowed to occupy by the electron of that material. It is equal to the energy difference between the minimum energy of conduction band and the maximum energy of valence band of that material. But in a bandgap the added impurity atoms can have their energy levels.

4.Distinguish Between direct bandgap and indirect semiconductors.

In direct bandgap semiconductors the electron from the conduction band can directly recombine with the hole in the valence band emitting a light photon and the charge carriers have smaller life time. Examples: Ga, As, In P. But in indirect bandgap semiconductors, the electron from the conduction band can recombine with a hole in the valence band in an indirect manner through the traps. The life time of charge carriers is more. Examples: Si, Ge.State Hall Effect.

5.Define Hall effect.

When a magnetic field is applied perpendicular to a current carrying conductor or semiconductor, a voltage is developed across the specimen in a direction perpendicular to both the current and the magnetic field. This phenomenon is called the Hall Effect and the voltage so developed is called the Hall Voltage.

6.Mention few applications of Hall Effect

1. The sign (electrons or holes) of charge carrier is determined.
2. The carrier concentration (number of charge carriers per unit volume) can be determined.
3. The mobility of charge carriers is measured directly.
4. Electrical conductivity of the material can be determined.

7.What is Hall Effect? What is its use in the semiconductors?

Hall Effect refers to the creation of a transverse e.m.f across the semiconductor slab carrying current in the perpendicular magnetic field. Using this effect the concentration and the sign of charge carriers can be determined. Further the mobility of charge carriers can also be determined.

8.What do u meant by Superconductivity and Supercondutor?

The ability of certain ultra cold substances to conduct electricity without resistance is called superconductivity. Substances having this property are called superconductors. A superconductor is a conductor having almost zero resistivity and also behaves as diamagnetic below its superconducting transition temperature.

9.Define Transition temperature.

The superconducting transition temperature 'Tc' of a material is defined as a critical temperature at which the resistivity of the material is suddenly changed to zero. Thus at that temperature of a material is changed from normal material to superconductor.

What are the changes occurs in conducting material at transition temperature?

At the transition temperature, the following physical changes are observed.

1. The electrical resistivity drops to zero.
2. The magnetic flux lines are excluded from the material.
3. There is a discontinuous change in specific heat.
4. Further there are also small changes in thermal conductivity and the volume of the material.

10.State Meissner effect.

The superconductor is a perfect diamagnet. As the material which is placed in an uniform magnetic field(whose value is smaller than the critical magnetic field Hc), is cooled below Tc the magnetic flux inside the material is excluded from the material. This is called MEISSNER EFFECT.

Thus a material can behave as a superconductor only when

1) the resistivity of the material should be zero and
2) the magnetic induction in the material should be zero when it is placed in an uniform magnetic field.

11.What is called Type I Superconductor?

1. Type I Super conductor is the one which exhibits a complete Meissner effect or perfect diamagnetism. i.e., above the critical field Hc, the specimen is a normal conductor. Below the critical field, the specimen excludes all the magnetic lines of force inside the specimen or it will become a diamagnetic material.
2. There are also called soft superconductors.
3. The values of Hc are always too low for these materials.
4. Examples: Al, Zn, Ga.

12.What are Type II Superconductors?

1. Type II Superconductors are the superconductors in which the magnetic flux starts to penetrate the specimen at a field Hc1 which is lower than critical field Hc. The specimen is in a mixed state between Hc1 and Hc2 and it has

superconducting electrical properties upto Hc2. Above Hc2 the specimen is a normal conductor.

2. Incomplete Meissner effect occurs in the region between Hc1 and Hc2 and this region is called vortex region. In this region the magnetic flux lines gradually penetrate the specimen as the magnetic field is increased beyond Hc1.
3. Examples: Zr, Nb, 60% Nb-40% Ti alloy.